



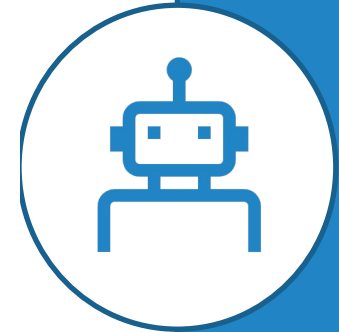
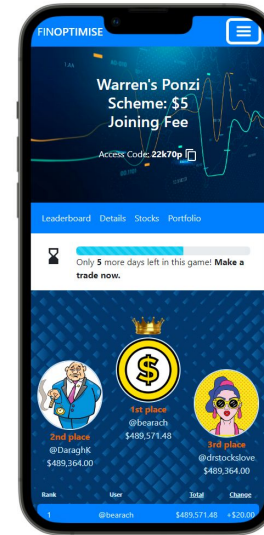
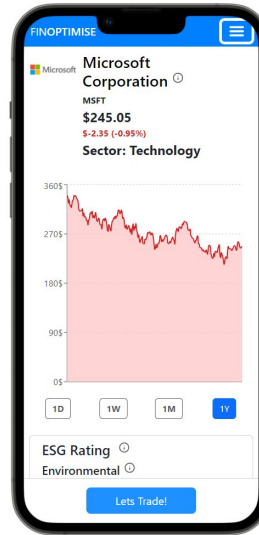
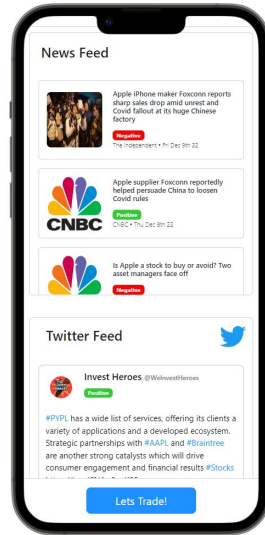
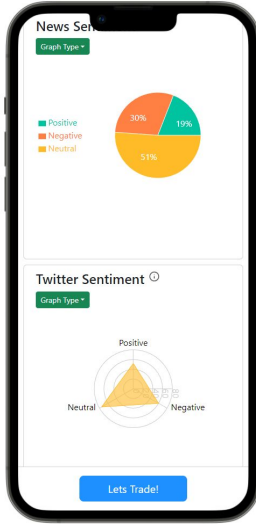
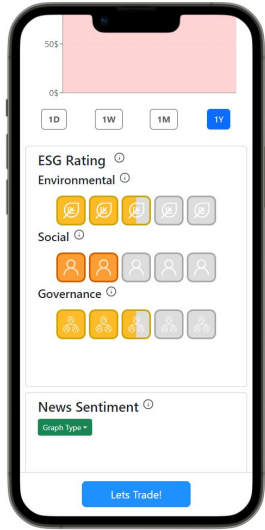
18th annual International
Technology, Education and
Development Conference

Valencia (Spain) - 4th, 5th
and 6th of March, 2024.

FinOptimise: A Comprehensive User Experience Methodology for Assessing a Stock Trading Simulation

CAOLAN POWER, DARAGH KNEESHAW,
REBECCA KELLY, WARREN KAVANAGH,
JOSEPH CORCORAN, BEARACH BYRNE,
ANDREA CURLEY, DAMIAN GORDON, PAUL
KELLY

Project Aim



“To provide a beginner-friendly platform allowing users to experience virtual investing and to provide a starting point for building a healthy interest in investing”

Personas

- User One
 - Financial goals – Move out of parents' house
 - Investing knowledge – None
 - Tech literacy - Basic
-
- User Two
 - Financial goals – Save for a house
 - Investing knowledge – Limited
 - Tech literacy - Advanced


Name Warren Buffet
Job Title Sales assistant
Age 18 to 24 years
Highest Level of Education High school degree or equi


Name Michelle Murphy
Job Title Project Manager
Age 35 to 44 years
Highest Level of Education Bachelor's degree (e.g. BA,

- Justification

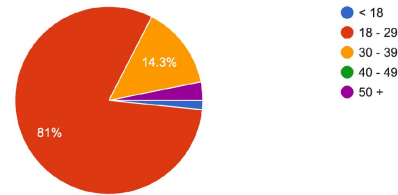
- Survey

- Vast majority in 18-39 age range
- Majority thought games were good for
- 44% really felt the cost-of-living crisis
- Majority wanted to invest ethically

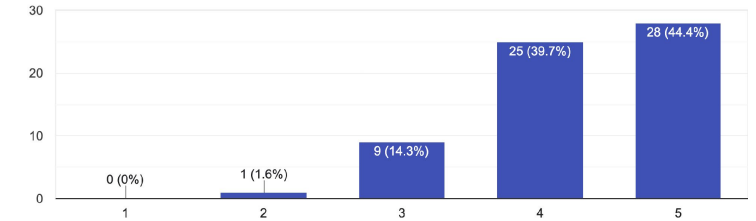
- Market Research

- Peak CPI this year October 9.2% - CSO
- The average stock market return over the last century - 10% - SnP500 index

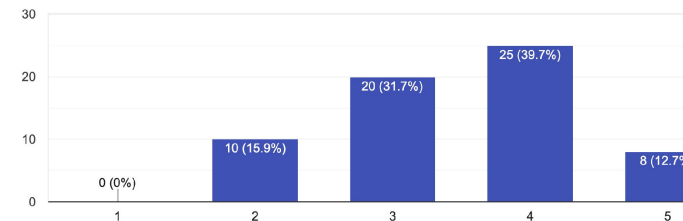
What age group are you in?
63 responses



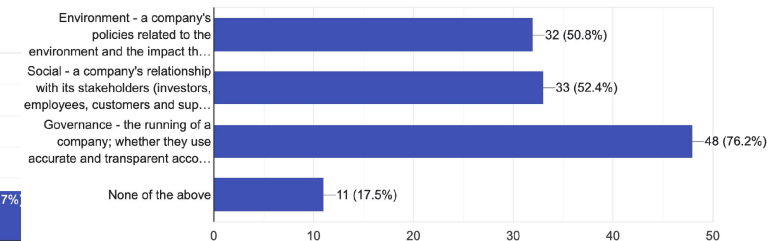
How effective do you think games are for learning?
63 responses



On a scale of 1 - 5, how much is the cost of living crisis affecting you?
63 responses



Would the following factors impact your decision to invest in a company or not?
63 responses



Technology Stack

React

- Component based
- Single Page App
- Familiarity

NodeJS

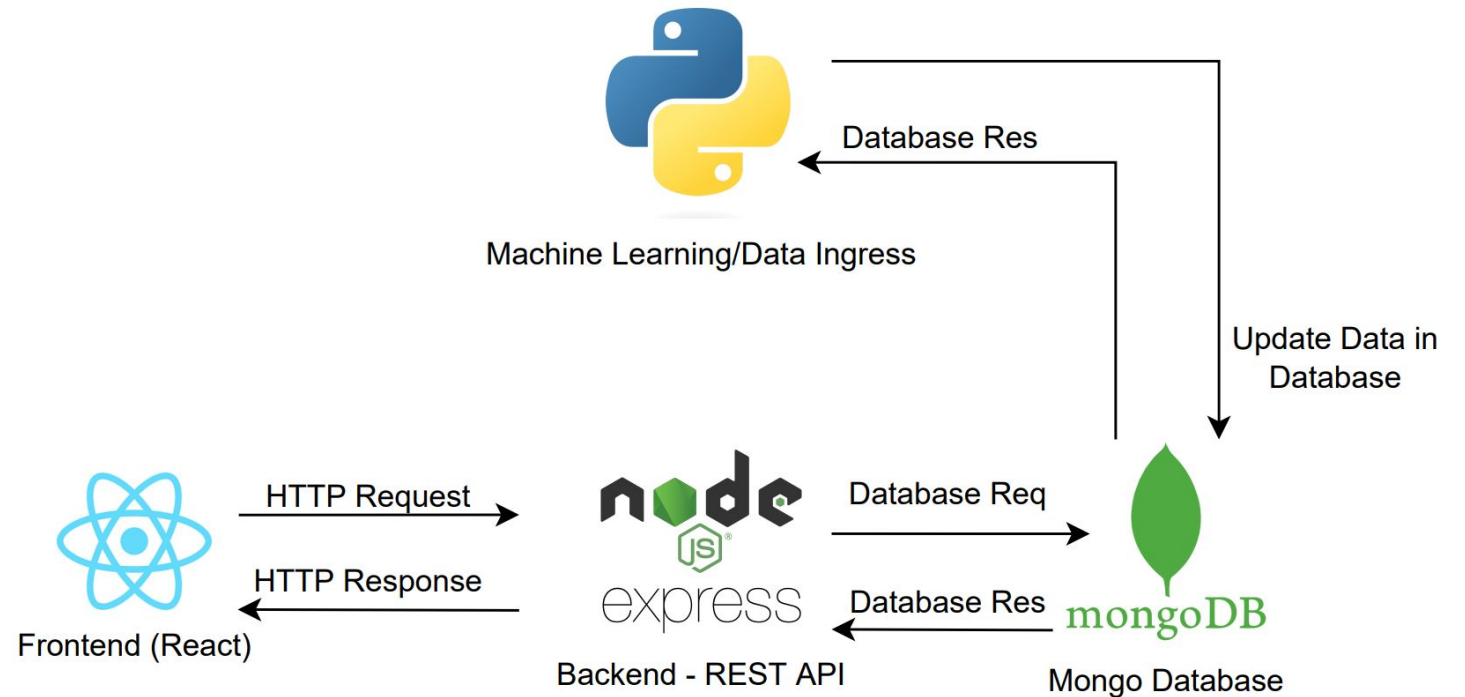
- Asynchronous Architecture
- NPM and Syntax
- Familiarity

MongoDB

- Schemaless Architecture
- Works well with JSON

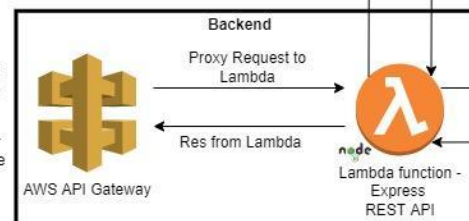
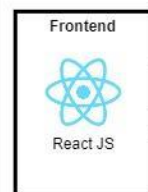
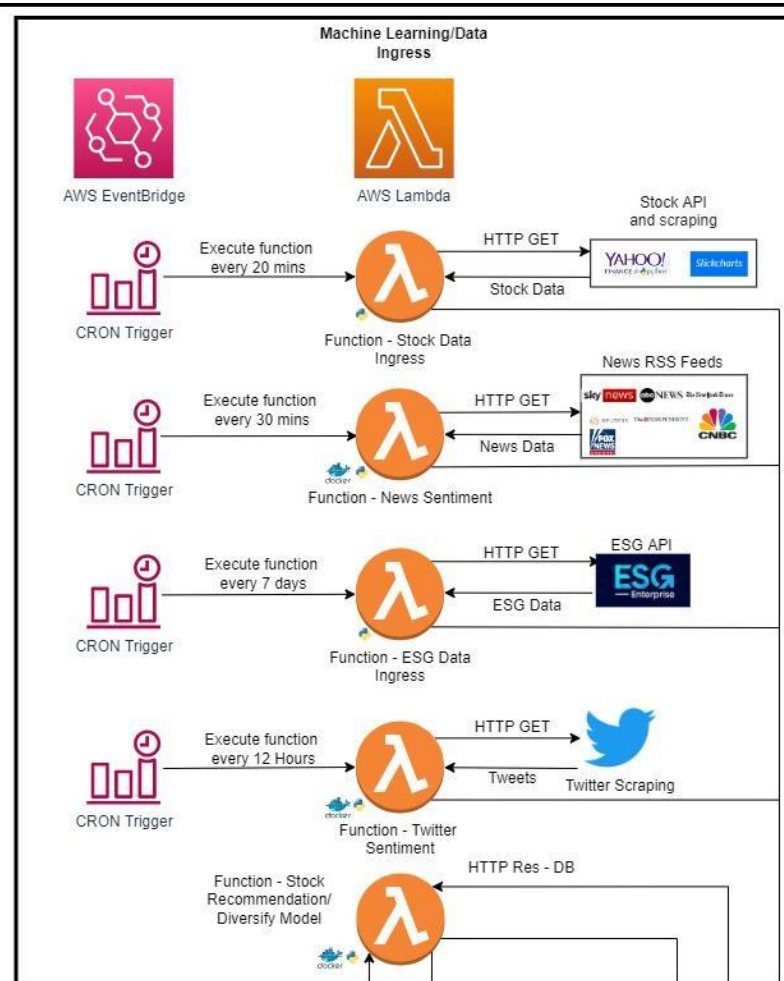
Python

- Data manipulation
- Better modules for data/ML





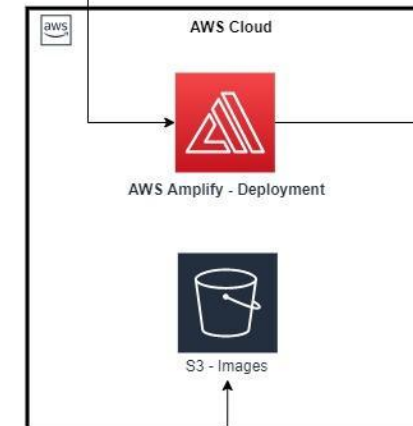
AWS Amplify - Project



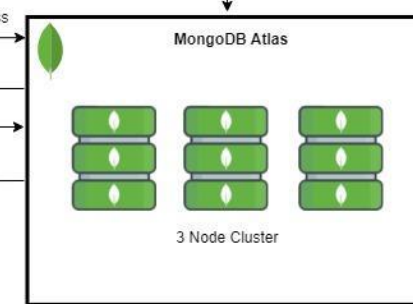
The AWS Amplify Project code gets pushed to GitHub



Master branch sent for deployment to AWS Amplify project on AWS Cloud



DB Stores Image URL

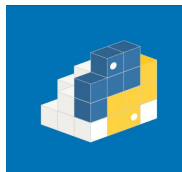


Data Resources

Data	Source	Collection Method	Storing Method	Accessing
Stock Data	Kaggle	One Time (.csv)	MongoDB	Mongoose
Live Stock Price Data	SlickCharts.com	Web Scraping (Lambda)		
Historical Stock Price Data	YFinance	One Time (Python Package)		
Stock ESG Data	ESG Enterprise API	API Calls (Lambda)		
News Data	Various News Sites (RSS)	API Calls (Lambda)		
Twitter Data	Twitter.com	Web Scraping (SN Scape)	.csv & .pkl files on Docker image (AWS SAM)	Express (Axios) Calls
Recommender Data	- Stock Data - User Transactions	Lambda (running as API through API Gateway)		

kaggle

Slickcharts



ESG
— Enterprise



sky
news

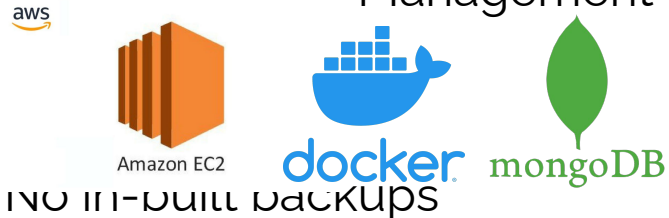
FOX
NEWS
channel



CNBC

Database Deployment

Initial Solution - AWS EC2 VM:

- **Pros:**
 - Cheap
 - Extremely Scalable
 - **Cons:**
 - Difficult to manage
 - Difficult Identity Management
 - NO IN-BUILT BACKUPS
- 

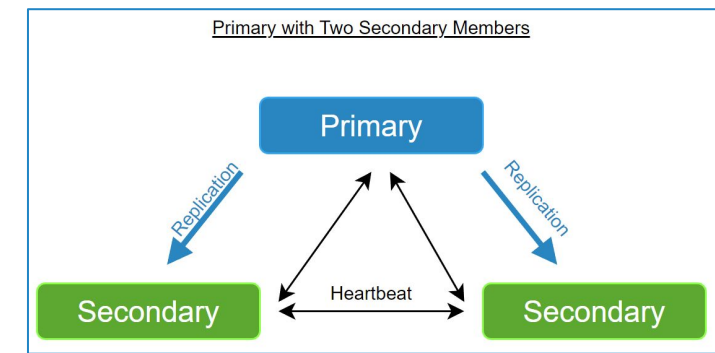
Updated Solution – MongoDB Atlas Dedicated:

- AWS M10 Cluster
- **Pros:**
 - Free - first \$50 (approx. 20 days)
 - 3 Node Replication
 - Easy Identity Management
- **Cons:**
 - Cost (without coupon codes)
- Triggers – Fundamental to Game Leagues
 - Start Game
 - Finish Game

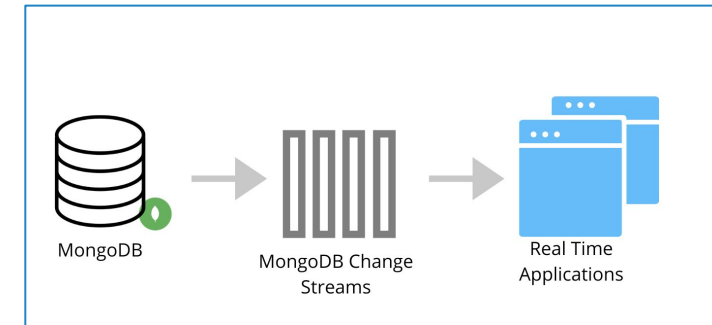


Database Details

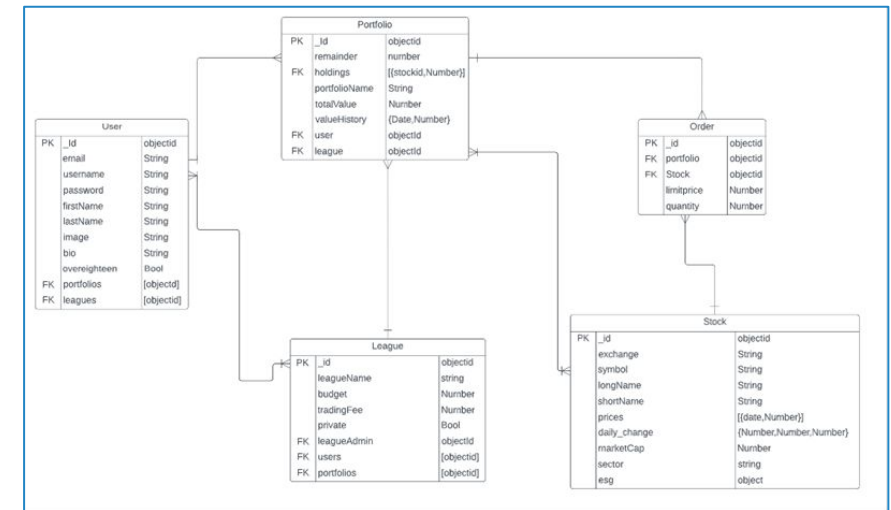
- Development & Production
- Identity Management
- Triggers – Fundamental to Game Leagues:
 - Start Game League
 - Finish Game
 - Value Based
 - Time Based
 - Update Portfolio History
 - Resetting Daily Trades for Active Portfolios



3 Node Replica Set



Mongo Triggers/Change Streams

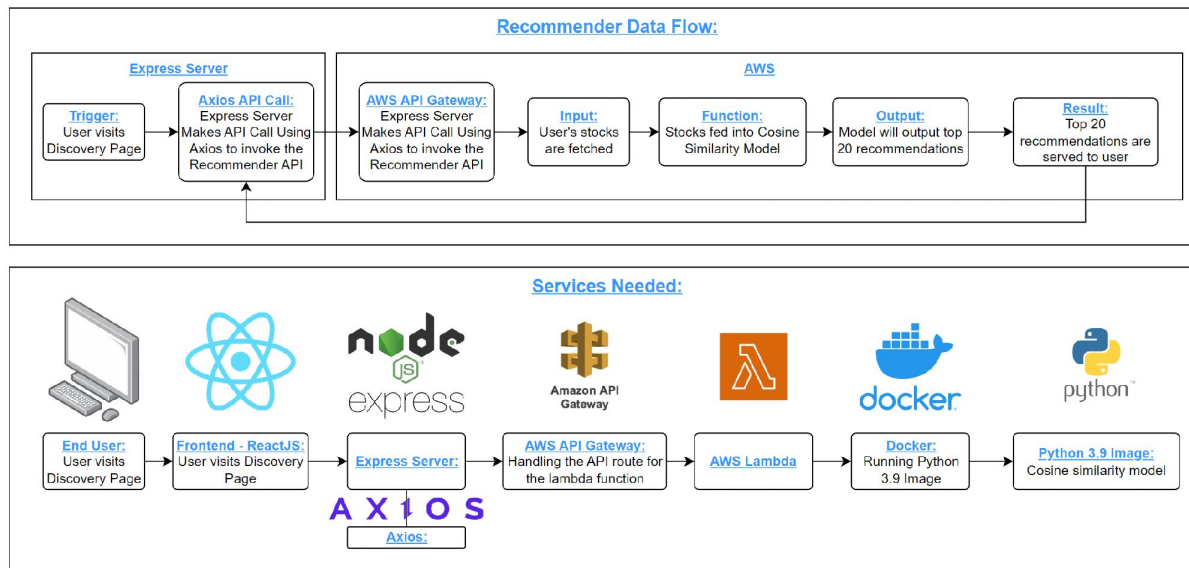


ERD

Machine Learning

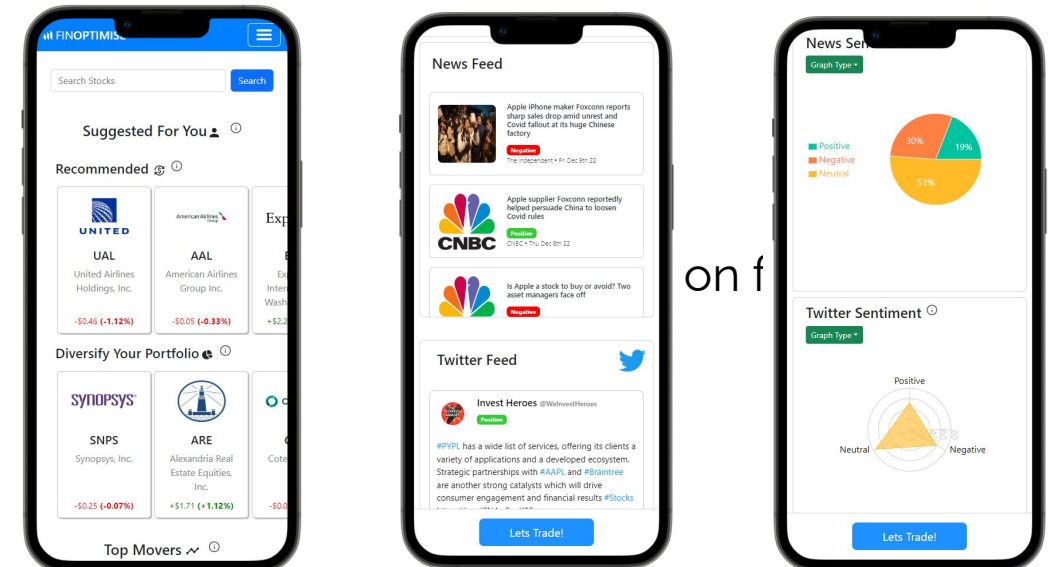
Recommender System:

- Content Based Recommender
- Cosine Similarity Model
- User transactions used as input.
- 2 types of recommendations:
 - Stocks **similar** to user's holdings



Sentiment Analysis:

- 2 Systems at Work:
 - News Sentiment
 - Twitter Sentiment
- Data Collection
- Data Cleaning



Phases of user evaluation



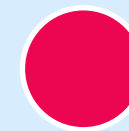
Phase 0
Market Research



Phase 2
Usability Testing

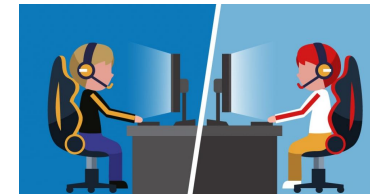
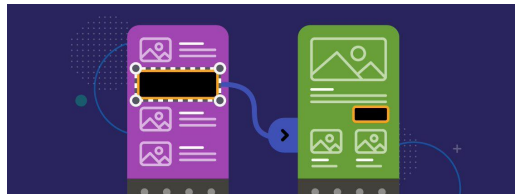


Phase 4
Usability Testing



Phase 1
Feature Prototyping

Phase 3
Autobiographical Testing



Questions per phase of evaluation

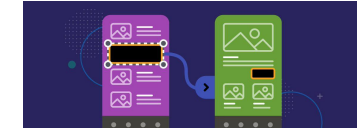
Phase 0: Market research and Market Analysis

- Survey
- Comparisons of similar systems



Phase 1: Feature Prototyping

- Think Aloud



Phase 2: End to End Usability testing

- Cognitive Walkthroughs
- Expert Reviews



Phase 3: Auto Biographical testing

- Continuous playing while developing by members of the team



Phase 4: End to End Usability testing #2

- Cognitive Walkthroughs



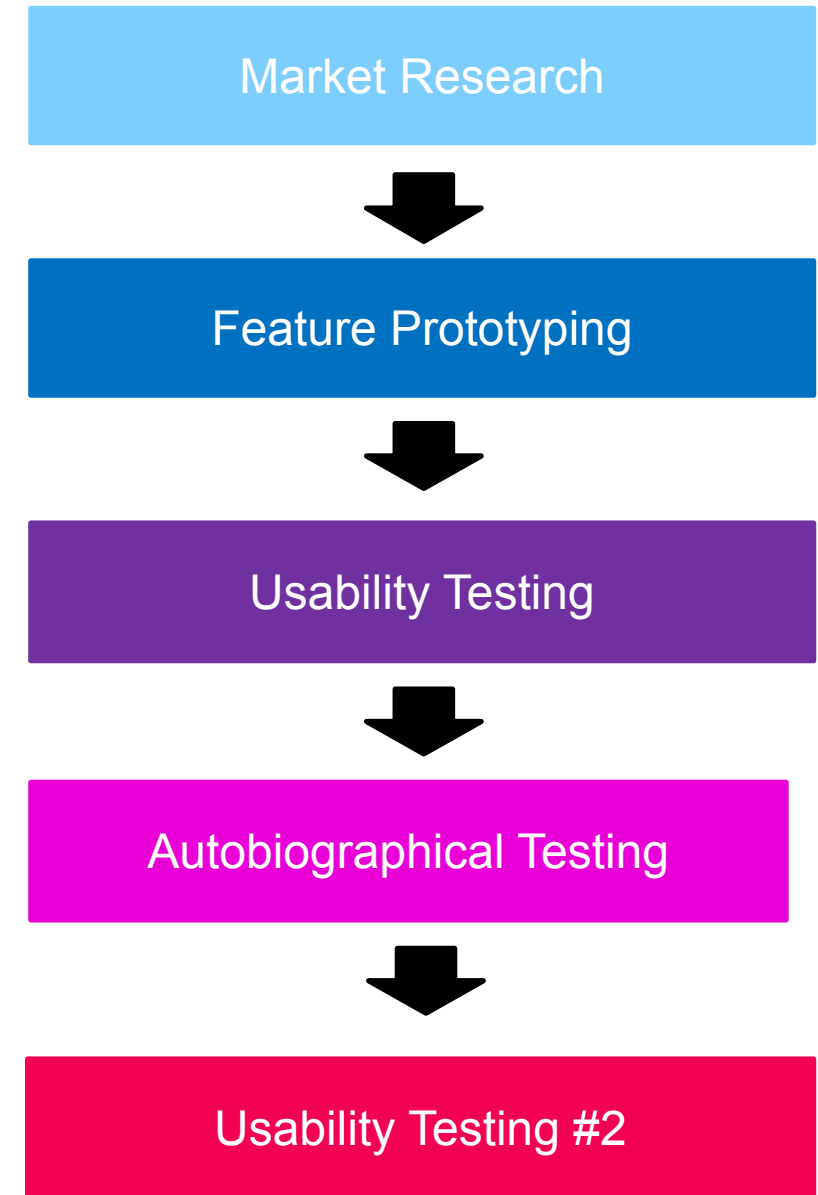
Data collection and analysis

- ▶ **Qualitative** data was all collated and each opinion was considered, as the amount of subjects used in each phase was low
- ▶ **Quantitative** data was analysed in terms of percentages (i.e. pass or fail of a task)



Conclusions

- ▶ **Phase 0 – Market Research (Survey)**
 - Established a target user group
- ▶ **Phase 1 – Feature Prototyping (Think Aloud)**
 - Iterative Process
 - Discovered things we didn't think about
 - Design based on users
- ▶ **Phase 2 – Usability Testing (Cog Walk & Expert User)**
 - Tested usability of the app
 - Pointed out areas in app we have missed
 - App does not feel like a game
 - A lot of usability issues
- ▶ **Phase 3 – Autobiographical Testing**
 - Tested playability of the game
 - Continuously updated game
- ▶ **Phase 4 – Usability Testing #2 (Expert user & UX class)**
 - Tested usability of the app
 - Users performing tasks



Team-Work

Positives

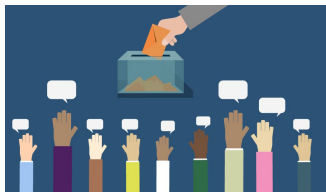
- Constant Communication



- Good planning

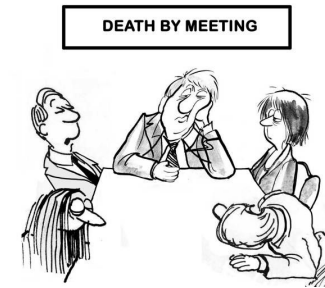


- Democratic process



Negatives

- Meetings too long



- Too many people on calls



- Perfect features vs add new features

What we would do differently

- Project Management tool



- Allow more time for tasks



- Concentrate on one task at a time



- Think about Database choice more





Q&A